

Facial Expressivity Features and Biomarkers for PTSD, Depression, and Anxiety Disorders

Executive summary

The strongest and most mature facial-analysis literature in the retrieved primary sources is for **depression**. Across challenge datasets and newer naturalistic studies, depression-related facial markers are most consistently described as **reduced positive expressivity, lower overall facial variability, smoother and less reactive temporal dynamics, reduced head movement**, and, in some studies, altered **action-unit patterns** such as less affiliative activity in **AU12** and **AU15** and more non-affiliative **AU14**. These signals are usually operationalized through facial landmarks, automated FACS/AU coding, head-pose trajectories, gaze direction, and multi-scale temporal modeling rather than through single static frames alone. ¹

The evidence for **anxiety disorders**, especially **social anxiety**, is smaller but growing. In the retrieved papers, the most promising facially observable markers are **gaze behavior, head position/pose, facial landmarks**, and selected **AUs** recorded during social interaction. The best-performing recent study in an unstaged social setting reported very strong classification accuracy from head-position, landmark, eye-movement, and AU features, but the literature is still thin, and external validation remains limited. Eye-tracking work suggests that attentional scan paths to faces may contain signal for both depression and social anxiety, but current performance is much weaker than the best face-video models. ²

For **PTSD**, the visual-facial evidence base is the sparsest of the three disorder groups. The most concrete retrieved work was a new unconstrained video dataset with a ResNet50v2+LSTM visual baseline; interpretability at the level of clinically meaningful facial biomarkers is still limited compared with depression. In practice, PTSD papers in this slice of the literature lean more toward **multimodal** or **audio/text-driven** detection, while facial-video work is at an earlier stage and is still establishing datasets, benchmarks, and robust cross-context features. ³

A core conclusion across disorders is that these are best treated as **candidate digital biomarkers for screening and monitoring**, not as standalone diagnostic biomarkers. The most credible path forward is an interpretable, calibrated, human-in-the-loop pipeline that combines **facial dynamics, speech/language, clinical metadata**, and ideally **longitudinal within-person baselines** rather than one-off face-only classification. ⁴

Scope and evidence base

The two Emora documents referenced in the request were **not visible in the accessible conversation context**, so this report does **not** summarize them directly. The synthesis below is based on the research sources that were successfully surfaced and inspected.

The retrieved evidence is concentrated around a few benchmark families and a few newer naturalistic datasets. For depression, the most important sources here were **DAIC-WOZ** and **AVEC 2013/2014/2017/2019**, plus more recent naturalistic work such as **FacePsy**, **MoodCapture**, and a large adolescent dataset of **11,427** facial videos. For PTSD, the clearest facial-video source retrieved was **PTSD in the Wild**, whose published train/validation/test counts imply **317 PTSD** and **317 non-PTSD** videos in total. For anxiety, the clearest facially grounded source was **AnxietyFaceTrack** with **91 participants**, plus a smaller eye-tracking paper on depression and social anxiety. ⁵

A notable methodological pattern is that explicit **microexpression-only** pipelines were uncommon in the primary disorder-specific papers retrieved here. The dominant successful representations were **AUs**, **landmarks**, **head motion/pose**, **gaze/eye measures**, and especially **temporal dynamics** modeled across short slices and/or full interviews. That is a useful practical takeaway: in this literature, coarse-to-meso temporal structure currently appears more actionable than rare, high-speed microexpression detection. ⁶

Feature families and their clinical interpretation

A central distinction in this literature is between a **signal that is easy to compute** and a **signal that is clinically interpretable**. Facial expressivity is often used as a proxy for constructs such as social withdrawal, anhedonia, psychomotor slowing, emotional blunting, hypervigilance, avoidance, or social-attentional bias. Importantly, the same computer-vision variable can mean different things depending on context. For example, “less smiling” can reflect reduced positive affect, interpersonal disengagement, suppression, cultural display rules, task mismatch, or medication effects. Several papers therefore emphasize task context, temporal structure, and multimodal corroboration instead of treating any single facial feature as disorder-specific. ⁷

The feature-to-clinical map below stays close to the retrieved primary sources and explicitly marks under-specified areas.

Feature family	How it is measured with computer vision	What it is intended to measure clinically or psychologically	Evidence in retrieved literature
Reduced positive expressivity	AU coding, especially smile-related movements such as AU12 ; emotion probabilities; smile counts/duration/intensity from frame or video models	Often interpreted as anhedonia , reduced affiliative signaling, reduced positive affect display, or social withdrawal	Depression papers repeatedly describe reduced positive facial displays; one review section specifically summarizes fewer affiliative AU12/AU15 displays in higher depression severity. FacePsy also found smile expressions and AUs including 6, 12, 15, 17 informative. ⁸

Feature family	How it is measured with computer vision	What it is intended to measure clinically or psychologically	Evidence in retrieved literature
Non-affiliative / negative facial actions	AU presence/ intensity, especially AU14 and sadness-related AUs/emotion outputs	Often intended to reflect interpersonal distancing, negative valence, or tension-related expression	The depression literature summary in Xu et al. reports more AU14 with higher depression severity; recent AU-focused exploratory work found group differences in sadness/happiness-related AUs. ⁹
Head motion and head pose	Yaw/pitch/roll trajectories, motion-energy features, “kinemes,” landmark-derived head displacement	Often interpreted as psychomotor slowing, reduced engagement, downward orientation, avoidance, or low social vigor	Retrieved depression papers repeatedly note reduced head motion; a head-motion-only study found kinemes useful for depression detection, and AnxietyFaceTrack found head position among the strongest individual feature sets for social anxiety. ¹⁰
Gaze direction and social attention	Eye-gaze estimation, gaze orientation, scanpaths, gaze density maps, gaze diversion frequency, downward gaze measures	Intended to capture avoidance, social-attentional bias, hypervigilance, reduced eye contact, or altered face processing	Haque et al. cite downward gaze as a clinical nonverbal cue; Faces of the Mind identified subtle shifts in gaze orientation; the eye-tracking paper frames both depression and social anxiety as involving face-directed attentional biases. ¹¹
Eye-open states, blinking, eyelid dynamics	Eye-aspect ratio, eye-open state estimation, blink counts/rates, eyelid landmarks	Intended to reflect fatigue, arousal, vigilance, suppression, or stress-related eye behavior	This is less consistently used than landmarks/AUs/head pose in the surfaced papers, but FacePsy reported eye-open states as significant. Specific blink-rate biomarker evidence was not well specified in the retrieved disorder papers. ¹²
Landmark dynamics and geometry	68-point or similar landmarks, distances, velocities, accelerations, affine normalization, Kalman filtering, jitter correction	Used as a compact proxy for facial-muscle movement and geometry while suppressing background noise	FacialPulse relies on 68 landmarks plus calibration/jitter reduction; AnxietyFaceTrack found landmarks highly predictive; many classical systems also use landmark or geometry-based video encoding. ¹³

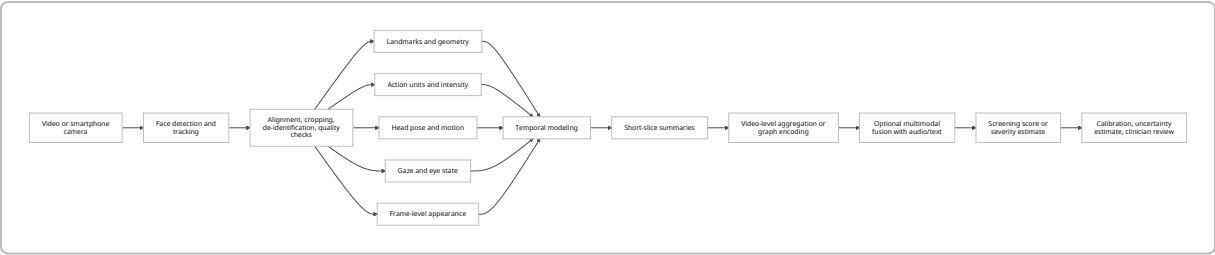
Feature family	How it is measured with computer vision	What it is intended to measure clinically or psychologically	Evidence in retrieved literature
Temporal smoothness, variability, and reactivity	Thin-slice models, optical-flow-like motion cues, temporal convolutions, BiGRU/LSTM, graph encodings, rank pooling, video-level trajectory statistics	Intended to capture psychomotor slowing, blunted reactivity, persistence of affect, and separation of transient emotion from sustained disorder-related behavior	This is one of the strongest recurring themes: FacialPulse explicitly shows smoother AU12/AU15 curves in depression; two-stage graph models argue that both short- and long-term temporal scales matter; topic-aware systems preserve temporal context instead of averaging whole interviews. ¹⁴
Pupil dynamics and fine ocular behavior	Pupil size/dynamics, gaze tracking, eye-region time series	Intended to index arousal, attention, and subtle mood-related shifts not obvious in gross expression	The large adolescent dataset specifically highlighted pupil dynamics and gaze orientation as potential mood-disorder markers. ¹⁵
Symmetry and asymmetry	Left-right AU intensity comparisons, facial landmark asymmetry, regional imbalance	Potentially intended to reflect atypical muscle recruitment or regulation; in this disorder set it remains more a proposed than established marker	In the surfaced disorder-specific papers, symmetry/asymmetry was not a major validated feature family; details were usually unspecified. This remains an open area rather than a high-confidence biomarker from the retrieved evidence. ¹⁶

Computer-vision measurement pipeline

Most successful systems in this literature follow a similar pattern: detect and track the face, normalize the geometry, extract either interpretable facial descriptors or learned representations, model **temporal structure**, optionally fuse with audio/text, and then predict either a diagnosis proxy or a severity score such as **PHQ-8/9** or **BDI-II**. The older hand-crafted pipeline was built around features such as **LBP-TOP**, **EOH**, **LPQ**, **OpenFace AUs**, and summarized interview statistics, while newer systems increasingly use **CNN-RNN**, **3D CNN**, **BiGRU**, **graph neural**, or **transformer-like** temporal encoders. Yet several recent papers still reach their strongest practical conclusions with **interpretable facial cues** plus classical ML, especially when the target setting changes across contexts. ¹⁷

Preprocessing choices matter more than they may appear. DAIC-WOZ uses low-resolution 3D face scans that were deliberately constrained for privacy while still preserving eye, lip, and head motion. Topic-modeling work showed that **context-aware segmentation** preserves informative local dynamics that are lost when features are averaged over an entire interview. In PTSD in the Wild, the visual baseline required manual extraction of interviewee frames, then MTCNN face cropping, and only **60 face-containing frames per video** were used for the visual model. FacialPulse explicitly adds landmark calibration to reduce

accumulated tracking error and jitter, showing that measurement stability is itself part of the biomarker problem. 18



If you are preparing this as a formal brief, the most useful figures to request from primary sources are: **FacialPulse Figure 2** for a clean landmark-plus-temporal pipeline; **Gong & Poellabauer Figure 1** for topic-aware context segmentation; **Xu et al. Figure 1** for a two-stage short-term/long-term graph model; and **PTSD in the Wild Figure 4** for the visual CNN-RNN baseline. Those figures are especially helpful because they show where the field has converged architecturally. 19

Comparative evidence from key studies

The table below compares the key studies surfaced in this search. Where the accessible abstract or inspected lines did not specify a requested detail, I mark it explicitly as **not specified**.

Citation	Disorder	Sample size	Features used	CV methods	Model	Metrics and results
Haque et al., 2018 20	Depression	189 interviews / 142 patients in DAIC-WOZ	3D facial scans, audio, text	3D facial keypoints/ scans; multimodal sentence embeddings	Causal CNN over audio + visual + text	PHQ error 3.67; 83.3% sensitivity, 82.6% specificity for MDD-style thresholding

Citation	Disorder	Sample size	Features used	CV methods	Model	Metrics and results
Gong & Poellabauer, 2018 ²¹	Depression	189 subjects (DAIC-WOZ / AVEC context)	Topic-aware audio, video, semantic features; OpenFace AUs in selected contexts	Topic segmentation; context-aware feature slots; feature selection	Regression with selected topic-conditioned features	Proposed method: RMSE 4.99, MAE 3.96, F1 0.60 on test; beat challenge/context-unaware baselines
Xu et al., 2021 ²²	Depression	AVEC 2013/2014 challenge sets; exact N not fully specified in inspected lines	Multi-scale short-term facial behavior features	Thin-slice modeling + depression feature enhancement + graph encoding	Two-stage deep model with SEG/SPG graph representations and GNN-style video modeling	Best reported on AVEC 2014 combined setting: MAE 5.86, RMSE 7.18 ; on AVEC 2013 SPG MAE 5.95, RMSE 7.57
FacialPulse, 2024 ²³	Depression	AVEC2014: 300 videos ; MMDA internal dataset size not specified in inspected lines	68 facial landmarks; temporal facial motion; landmark calibration	Face detection, alignment, landmark calibration, temporal trajectories	BiGRU-based temporal model with landmark-calibration module	AVEC2014: RMSE 7.60, MAE 5.92 ; MMDA baseline MAE improved from 4.35 to 3.97
Gahalawat et al., 2023 ²⁴	Depression	BlackDog + AVEC2013; exact N not specified in abstract	Head-motion “kinemes” only	Head-motion mining and pattern discovery	Classical ML on kineme features / reconstruction-error statistics	Peak F1 0.79 (BlackDog thin slices), 0.82 (AVEC2013 thin slices), 0.72 (AVEC2013 video-level)

Citation	Disorder	Sample size	Features used	CV methods	Model	Metrics and results
FacePsy, 2024 ¹²	Depression	25 participants	Eye-open states, head gestures, smile expressions, AUs 2, 6, 7, 12, 15, 17	Mobile affective sensing on smartphone; facial behavior and head-gesture extraction in naturalistic use	Screening classifier + PHQ regression	AUROC 81% ; PHQ-9 regression MAE 3.08
Faces of the Mind, 2024 ¹⁵	Depression + anxiety + stress	11,427 adolescents	Facial features from standardized videos; pupil dynamics; gaze orientation	Large-scale facial video analysis; clustering; tree-based and deep models	Tree models and deep learning	Main conclusion was not a headline clinical AUC but that small-dataset methods deteriorated at scale; identified pupil dynamics and gaze orientation as candidate markers
Pariikh et al., 2024 ²⁵	Depression	Sample size not specified in abstract	AU intensities and temporal AU series	AU extraction, statistical comparison, time-series classification, PCA/ clustering	Time-series models + exploratory unsupervised analysis	Significant group differences in AUs related to sadness and happiness; exact classification metrics not specified in abstract

Citation	Disorder	Sample size	Features used	CV methods	Model	Metrics and results
AnxietyFaceTrack, 2025 ²⁶	Social anxiety	91 participants	Eye movements, head position, facial landmarks, AUs	Smartphone video analysis in unstaged social interaction	Random Forest using top 20% features	91.0% multiclass accuracy; 92.33% average binary accuracy; head position and landmarks were strongest single-region features
Chlasta et al., 2025 ²⁷	Depression + social anxiety	Sample sizes of the two source studies not specified in abstract	Eye-gaze scan paths to faces	Eye tracking; scanpath images	ResNet-style CNNs	About 48% average accuracy for 3-class screening and 62% for 2-class screening
PTSD in the Wild, 2022 ²⁸	PTSD	634 total videos inferred from published split counts (317 PTSD / 317 non-PTSD)	Face video frames (60 frames/video for visual baseline), plus audio/text baselines	MTCNN face extraction; frame selection; CNN-RNN visual baseline	ResNet50v2 + LSTM for visual baseline	Visual baseline test accuracy 0.82, precision 0.82, recall 0.82, F1 0.82 ; 3-fold visual average 0.84 accuracy/F1

Across these studies, the broad pattern is consistent: **face-only depression systems can work reasonably well**, especially when temporal modeling is strong; **head motion and gaze** are valuable complementary

cues; **anxiety** is not yet well standardized; and **PTSD** still lacks a mature, interpretable facial-biomarker literature comparable to depression. ²⁹

Limitations, confounds, and ethics

The most important scientific limitation is **construct ambiguity**. Facial behavior is influenced by many factors besides psychiatric state: task demands, personality, social context, fatigue, camera position, language, cultural display rules, depression-anxiety comorbidity, and basic discussion topic. The topic-modeling paper made this point directly by showing that the same observable feature, such as smiling, can have different meaning depending on conversational context. The Haque paper also stressed that avatar-based interviews differ from human-human interviews, and the 2026 cross-context comparison found that both classical and deep visual approaches generalized only modestly across contexts. ³⁰

The second major limitation is **dataset design**. Many benchmark studies are small or challenge-driven, some rely on interview labels rather than frame-accurate symptom labels, and several accessible summaries do not specify medication status, treatment phase, or comorbidity controls. For PTSD, the face-video literature retrieved here is still benchmark-building rather than biomarker-mature. For anxiety, the best recent paper is promising but still based on one social setting and self-report-derived classes. This means the field remains vulnerable to overfitting, site effects, and optimistic metrics. ³¹

The third limitation is **technical confounding**. Lighting, background clutter, occlusion, focus, pose, and resolution are explicitly called out in PTSD in the Wild. FacialPulse shows that even landmark jitter and cumulative tracking error materially affect downstream results, which is a reminder that “biomarker” quality depends on measurement quality. In mobile and in-the-wild studies, device placement, user behavior, and sampling frequency can become hidden nuisance variables. ³²

On ethics and privacy, these systems are sensitive even when they are technically de-identified. DAIC-WOZ used low-resolution 3D face scans precisely to reduce identifiability while preserving motion, and MoodCapture explicitly examined user concerns around sharing smartphone images for depression screening. PTSD in the Wild also highlights that mental-health AI raises serious privacy and stigma risks and that dataset scarcity often collides with the need to protect vulnerable populations. In practice, face-based psychiatric screening should be treated as **high-risk inference**: it requires informed consent, strong data minimization, secure storage, clear retention limits, and human review for any consequential use. ³³

Recommended improvements and implementation notes

The most concrete improvement is to move away from monolithic “black-box face diagnosis” toward **interpretable temporal feature sets**. In this literature, interpretable features with the strongest face validity are AU trajectories, landmark dynamics, head motion, gaze orientation, and eye-state variables, all modeled over time. A practical system should log not just a final score but also **what changed**: smile rate, AU12/AU15 amplitude, head-down dwell time, gaze aversion bursts, landmark variability, and within-session reactivity to specific clinically meaningful prompts. The retrieved papers strongly support **temporal** and **context-aware** modeling over single-image classification. ³⁴

The second improvement is **multimodal fusion**, but with disciplined role separation. The PTSD paper showed text outperforming visual baselines; Haque’s DAIC-WOZ system benefited from audio + visual +

language; and several depression papers show that face-only systems remain informative but incomplete. A robust clinical screening stack should therefore treat facial expressivity as one channel among several: facial behavior for nonverbal expressivity, speech for psychomotor slowing and prosody, language for content and cognition, and questionnaires for symptom anchoring. ³⁵

The third improvement is **longitudinal within-person monitoring** rather than one-shot case-control classification. Haque et al. explicitly identified longitudinal modeling as future work, and this recommendation follows logically from the scale and generalization results in the newer literature. For deployment, the most clinically meaningful baseline may be “this person is less expressive / more avoidant / more slowed than *their own* historical norm,” not “this face resembles a depressed class average.” That is especially important for PTSD and anxiety, where state, context, and trigger exposure matter greatly. ³⁶

The fourth improvement is **domain adaptation and fairness testing**. The 2026 cross-context comparison is a warning that performance and fairness can shift across interaction contexts, and the large-scale adolescent paper is a warning that coefficients learned on small datasets may not survive at scale. Any practical implementation should therefore evaluate by site, camera, culture/language group, sex/gender, age band, and task type, and should include calibration curves rather than only discrimination metrics. ³⁷

The fifth improvement is **calibration and uncertainty estimation**. Many papers optimize RMSE, MAE, F1, or accuracy, but practical screening needs probability calibration, abstention thresholds, and uncertainty flags for poor lighting, occlusion, or out-of-distribution behavior. A usable product should output something like: “screen positive with moderate confidence; reliability reduced because face visibility was poor.” This is an inference from the measurement-quality issues explicitly described in the retrieved dataset and modeling papers. ³⁸

Finally, for implementation, the least fragile starting point appears to be:

1. **Face quality gate**: reject frames with low visibility, poor illumination, or extreme occlusion.
2. **Interpretable feature extraction**: landmarks, AUs, head pose, gaze, eye-state.
3. **Temporal aggregation**: short slices plus session-level summaries.
4. **Optional multimodal fusion**: add audio/text only after the visual subsystem is stable.
5. **Calibrated output**: probability plus uncertainty, not only a hard label.
6. **Human oversight**: use for triage/monitoring, not autonomous diagnosis.

That recommendation is a synthesis of the highest-confidence findings in the retrieved sources rather than a single paper’s prescription. ³⁹

Open questions and explicitly under-specified details

Several details requested in the prompt were **not consistently specified** in the accessible source views. These include medication status, structured comorbidity controls, and in some cases exact sample sizes for older or abstract-only papers. Symmetry/asymmetry, blink-rate biomarkers, and microexpression-specific pipelines were also **not prominent** in the retrieved disorder-specific primary papers, so they should be treated as promising but under-validated directions rather than established facial biomarkers for PTSD, depression, or anxiety disorders. ⁴⁰

If this report is converted into a manuscript or slide deck, the highest-value next curation step would be to pull the original figures and full tables from the AVEC/DAIC-WOZ, FacialPulse, Xu et al., FacePsy, AnxietyFaceTrack, and PTSD in the Wild papers, because those materials would give you the cleanest visual evidence for feature extraction, temporal modeling, and the current performance ceiling in each disorder family. 41

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